

ASSIGNMENT-12

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BATCH: 30

CODE:

```
import hashlib
```

```
import time
```

```
# -----
```

```
# SMART CONTRACT SIMULATION
```

```
class SimpleStorage:
```

```
    def __init__(self):
```

```
        self.stored_value = 0
```

```
        self.transactions = []
```

```
    def set(self, value):
```

```
        timestamp = time.strftime("%Y-%m-%d %H:%M:%S")
```

```
        tx_hash = self.generate_hash(value, timestamp)
```

```
        self.stored_value = value
```

```
        transaction = {
```

```
            "value": value,
```

```
            "hash": tx_hash,
```

```
            "time": timestamp
```

```
        }
```

```
self.transactions.append(transaction)
```

```
return transaction
```

```
def get(self):
```

```
    return self.stored_value
```

```
def generate_hash(self, value, timestamp):
```

```
    data = f"{value}{timestamp}"
```

```
    return hashlib.sha256(data.encode()).hexdigest()
```

```
# -----
```

```
# CONTRACT OBJECT (DEPLOYMENT)
```

```
contract = SimpleStorage()
```

```
# -----
```

```
# GUI FUNCTIONS
```

```
def store_value():
```

```
    try:
```

```
        value = int(entry.get())
```

```
        tx = contract.set(value)
```

```
    status_label.config(
```

```
        text="Transaction Successful ✓",
```

```
        fg="green"
```

```
    )
```

```
    hash_label.config(
```

```
        text=f"Transaction Hash:\n{tx['hash']}"
```

```
    )
```

```

        update_history()

    except ValueError:
        status_label.config(text="Please enter a valid integer.", fg="red")

def get_value():
    value = contract.get()
    result_label.config(text=f"Stored Value: {value}")

def update_history():
    history_text = "Transaction History:\n"
    for tx in contract.transactions:
        history_text += f"{tx['time']}: Stored {tx['value']} (Hash: {tx['hash']})\n"
    history_label.config(text=history_text)

# -----
# GUI SETUP

from tkinter import *

root = Tk()

root.title("Simple Storage Contract Simulation")

entry = Entry(root, width=20)

entry.pack(pady=10)

store_button = Button(root, text="Store Value", command=store_value)

store_button.pack(pady=5)

get_button = Button(root, text="Get Stored Value", command=get_value)

get_button.pack(pady=5)

status_label = Label(root, text="")

status_label.pack(pady=5)

hash_label = Label(root, text="")

hash_label.pack(pady=5)

```

```
result_label = Label(root, text="")
```

```
result_label.pack(pady=5)
```

```
history_label = Label(root, text="Transaction History:")
```

```
history_label.pack(pady=10)
```

```
root.mainloop()
```



